

Exercise 2

Question 1

```
SELECT DISTINCT department  
FROM Students;
```

department
IT
HR
Finance

Question 2

```
SELECT department,  
       AVG(age) AS avg avg_age  
FROM Students  
GROUP BY department;
```

department	avg_age
IT	20,5
HR	22
Finance	23

Question 3

```
SELECT department,  
       student_count COUNT(*) AS student_count  
FROM Students  
GROUP BY department  
HAVING COUNT(*) > 1;
```

department	Student-Count
IT	2
HR	2

Question 4

```
SELECT Student-ID,
       name,
       age,
       department
FROM Students
WHERE age BETWEEN 21 AND 23;
```

Student-id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

Question 5

```
SELECT Student-id,
       name,
       age,
       department
FROM Students
WHERE department IN ('IT', 'HR') AND age > 21;
```

Student-id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Question 6

```
SELECT department,  
       SUM(credits) AS total_credits  
FROM courses  
GROUP BY department  
HAVING SUM(credits) > 5;
```

department	total_credits
IT	11

Question 7

```
SELECT course_id,  
       course_name,  
       department,  
       credits  
FROM courses  
WHERE credits > 4;
```

Course-id	Course-name	Department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

Question 8

```
SELECT course-id, course-name, credits  
FROM courses  
ORDER BY credits DESC  
LIMIT 3;
```


Course id	Course-Name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

Question 9

```

SELECT MAX(grade) AS max-grade,
       MIN(grade) AS min-grade,
       AVG(grade) AS avg-grade
FROM enrollments;

```

Max-grade	Min-grade	avg-grade
90	78	84.6

Question 10

```

SELECT course-id,
       COUNT(*) AS enrollment-count
FROM enrollments
GROUP BY course-id;

```

Course-id	enrollment-count
101	1
102	1
103	1
104	1
105	1

Question 11

```
SELECT department,  
       SUM(salary) AS total_salary  
       SUM(bonus) AS total_bonus  
FROM salaries  
GROUP BY department;
```

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

Question 12

```
SELECT department,  
       AVG(salary) AS avg_salary  
FROM salaries  
GROUP BY department  
HAVING AVG(salary) > 55 000;
```

department	avg_salary
IT	61 000
Finance	70 000

Question 13

```
SELECT employee_id,  
       name,  
       salary,  
       bonus,  
       (salary + bonus) AS total_compensation  
FROM salaries  
WHERE (salary + bonus) > 60 000;
```

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spire	70000	6000	76000
4	Tyre	62000	5500	67500

Question 14

```

SELECT department,
       SUM(budget) AS total_budget,
       AVG(budget) AS avg_budget
FROM Projects
GROUP BY department
HAVING AVG(budget) > 70000;

```

Department	total_budget	avg_budget
IT	270000	135000
Finance	80000	80000

Question 15

```

SELECT project_id,
       project_name,
       department,
       budget
FROM Projects
WHERE budget BETWEEN 50000 AND 120000
AND department <> 'Marketing';

```


Project-id	Project-name	department	budget
1	AI APP	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000